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**NAME: _____**

# VCE®SPECIALIST MATHEMATICS

## Units 3 & 4 Practice Written Examination 1

**Reading time: 15 minutes**
**Writing time: 1 hour**

## QUESTION AND ANSWER BOOK

### Structure of book

| *Number of questions* | *Number of questions to be answered* | *Number of marks* |
|---|---|---|
| 10 | 10 | 40 |

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are **NOT** permitted to bring into the examination room: any technology (calculators or software), notes of any kind, blank sheets of paper and/or correction fluid/tape.

**Materials supplied**
- Question and Answer Booklet of 14 pages.
- Formula Sheet.
- Working space is provided throughout the Question and Answer Booklet.

**Instructions**
- Write your **student name** in the space provided above on this page.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- All written responses must be in English.

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**Instructions**

Answer **all** questions in the space provided.

Unless otherwise specified, an **exact** answer is required to a question.

In questions where more than one mark is available, appropriate working **must** be shown.

Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

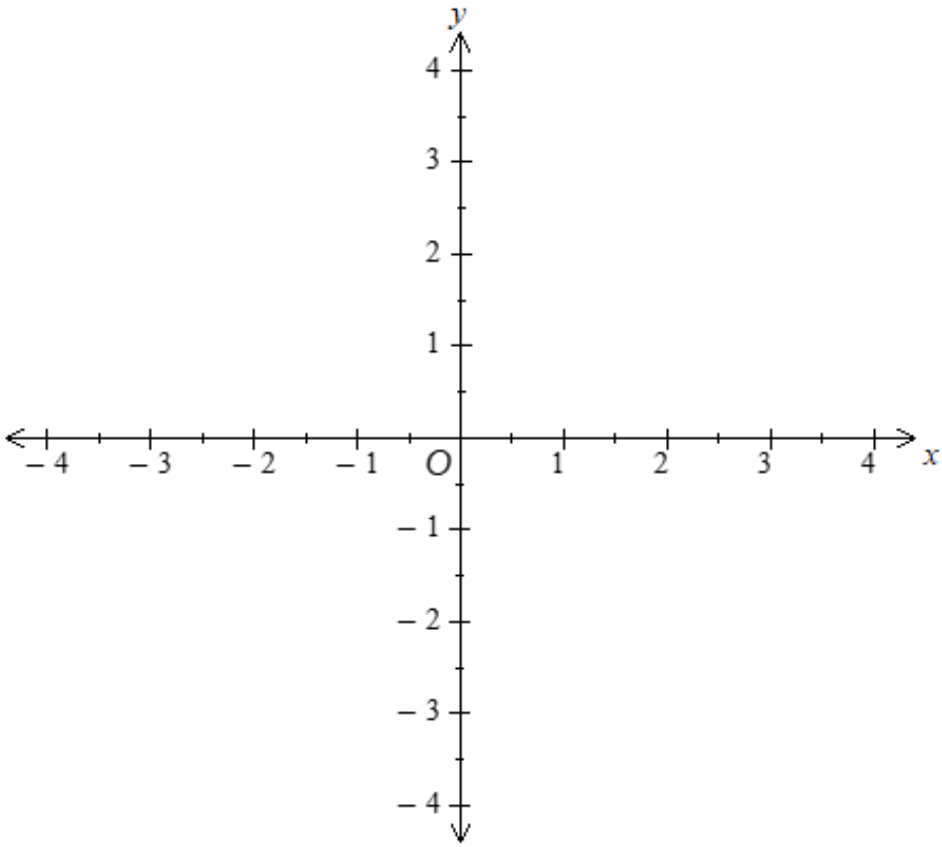
Take the **acceleration due to gravity** to have magnitude $g$ m$s^{-2}$ where $g = 9.8$.

**Question 1** (5 marks)

Consider the function $f_k$ with rule $f_k(x) = \dfrac{x^3 - x^2 + kx + 1}{x^2 + 1}$ where $k$ is a real constant.

**a.** Find the value(s) of $k$ for which the graph of $f_k$ does not intersect its asymptote. 2 marks

**b.** Sketch the graph of $f_k$ for $k = 3$ on the axes below. Label any asymptotes with their equation and any turning points and intersection with asymptotes with their coordinates. 3 marks




**Working space**

**Question 2** (4 marks)

Solve $3z^2 - 2\bar{z} = |2z - 1|$ where $z \in C$

**Question 3** (4 marks)

A particle with an initial velocity of 1 $ms^{-1}$ moves in a straight line. When the particle is $x$ m from its starting point its velocity, $v$ $ms^{-1}$, after $t$ seconds is given by

$$v = 2 - e^{-x}, \text{ where } x \geq 0.$$

Find the position of the particle when $t = \log_e(7)$.

**Question 4** (3 marks)

A particle moves in a straight line such that after $t$ seconds its displacement $x$ meters from a fixed origin $O$ is given by $t^3 + 2x^3 - xt = 2$.

Find the acceleration of the particle when $t = 1$ and $x = 1$.

**Question 5** (3 marks)

Evaluate $\int_{0}^{2\sqrt{3}} x\tan^{-1}\left(\frac{x}{2}\right)\,dx$

**Question 6** (4 marks)

The *Gratuitous Apple Farm* grows apples. It is known that the standard deviation of the mass of the apples is 5 grams. A random sample of 64 apples is used to calculate an approximate $C\%$ confidence interval for the mean mass, measured in grams, of all apples grown on the farm.

The confidence interval was found to be (201.2, 202.6).

**a.** $C = 100\Pr(|Z| < k)$ where $Z$ is the standard normal random variable.

Find the value of $k$. 1 mark

A larger sample of apples is selected, with a sample size four times the original sample. The sample mean is found to be the same.

**b.** State the approximate $C\%$ confidence interval for the mean mass, measured in grams, of all apples grown on the farm that would be calculated using this sample. Give all values correct to one decimal place. 1 mark

The *Gratuitous Apple Farm* also grows a small crop of genetically modified apples. The mass of these apples is normally distributed with a mean of 144 grams and a standard deviation of 8 grams.

**c.** Two genetically modified apples are chosen at random and compared to a third genetically modified apple. Find the probability that the combined mass of the two chosen apples is less than $\frac{7}{4}$ of the mass of the third apple. Use $\Pr(|Z| \leq 2) = 0.9545$ and give your answer correct to three decimal places. 2 marks

**Question 7** (4 marks)

The curve defined by the parametric equations

$$x = \cos^3(t), \quad y = \sin^3(t), \text{ where } 0 \le t \le \frac{\pi}{2}$$

is rotated about the *y*-axis.

Find the area of the surface generated.

**Question 8** (3 marks)

Use mathematical induction to prove that for $n \geq 2$,

$$\frac{(f_1(x)\cdots f_n(x))'}{f_1(x)\cdots f_n(x)} = \frac{f_1'(x)}{f_1(x)} + \ldots + \frac{f_n'(x)}{f_n(x)}$$

for all sets of differentiable functions $f_1(x)$, $f_2(x)$, $\ldots$, $f_n(x)$.

**Question 9** (5 marks)

Consider the following planes:

$$\Pi_1 : 3x - 2y - mz = 2$$

$$\Pi_2 : x + my + 3mz = 4$$

**a.** Find the value(s) of $m$ that will make $\Pi_1$ and $\Pi_2$ perpendicular to each other. 2 marks

**b.** Let $m = -1$.

Find a vector perpendicular to $\Pi_1$ and the line $x + 1 = \frac{2 - y}{2} = \frac{z - 3}{2}$ 3 marks

**Question 10** (5 marks)

The velocity vector of a particle at time $t$ seconds is given by

$$\underset{\sim}{v}(t) = \sin^2(t)\,\underset{\sim}{i} + \sin(t)\,\underset{\sim}{k}$$

The particle is initially at the point (0, 1, 0).

**a.** Find the position vector of the particle after $t$ seconds. 2 marks

The position vector of a second particle at time *t* seconds is given by

$$\underset{\sim}{s}(t) = \cos(2t)\underset{\sim}{i} - 3\underset{\sim}{j} + (2t - \sin(2t))\underset{\sim}{k}$$

**b.** Find the times at which the two particles are moving in the same direction. 3 marks

**END OF EXAMINATION**

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# VCE® Specialist Mathematics

## Practice Written Examination 1

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The VCAA formula sheet can be downloaded and printed from
https://www.vcaa.vic.edu.au/assessment/vce-assessment/past-examinations/Pages/Specialist-Mathematics.aspx

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## Solution Pathway

Below are sample answers. Please consider the merit of alternative responses.

**Specialist Mathematics Examination 1: Marking Scheme**

**Question 1**

**Part a.:**

A function can intersect with either a horizontal asymptote or a diagonal (oblique, slant) asymptote:

$$
\begin{array}{r|l}
 & x - 1 \\
\hline
x^2 + 1 & x^3 - x^2 + kx + 1 \\
 & -\left(x^3 + 0x^2 + x\right) \\
\hline
 & -x^2 + (k-1)x + 1 \\
 & -\left(-x^2 + 0x - 1\right) \\
\hline
 & (k-1)x + 2
\end{array}
$$

Therefore

$$f_k(x) = \frac{x^3 - x^2 + kx + 1}{x^2 + 1} = x - 1 + \frac{(k-1)x + 2}{x^2 + 1}$$

| | |
|---|---|
| **Diagonal asymptote:** $y = x - 1$ | **1 mark** |

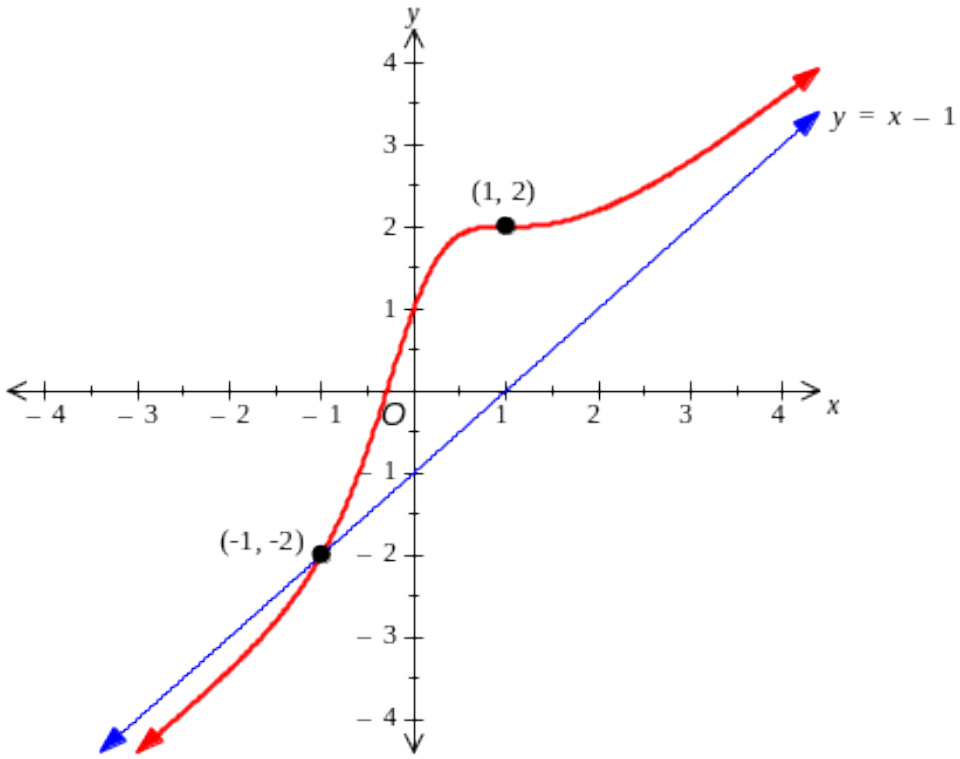
Solve $f_k(x) = x - 1 + \frac{(k-1)x + 2}{x^2 + 1} = x - 1$ : $\frac{(k-1)x + 2}{x^2 + 1} = 0 \Rightarrow (k-1)x + 2 = 0$.

By inspection there is no solution when $k = 1$

| | |
|---|---|
| **Answer:** $k = 1$ | **1 mark** |

**Part b.:**

**Answer:**




| | |
|---|---|
| **Shape:**<br>Must have a $y$-intercept at $(0,\ 1)$ | **1 mark** |
| **Turning point:** $(1,\ 2)$ | **1 mark** |
| **Diagonal asymptote:** $y = x - 1$<br>Graph must be consistent with axes scale.<br>**Intersection point:** $(-1,\ -2)$ | **1 mark** |

**Calculations:**

- Diagonal asymptote: $y = x - 1$ (from **part a.**).
- Intersection with asymptote:

Substitute $k = 3$ into $\frac{(k-1)x+2}{x^2+1} = 0$ from **part a.**: $2x + 2 = 0 \Rightarrow x = -1$ $y = f_3(-1) = -2$.

- Approach towards diagonal asymptote:

$y = x - 1 + \frac{2x+2}{x^2+1}$ (from **part a**.) $\sim x + 0^+$ as $x \to +\infty$

therefore, the approach is from above.

$y = x - 1 + \frac{2x+2}{x^2+1}$ (from **part a**.) $\sim x + 0^-$ as $x \to -\infty$

therefore, the approach is from below.

The approach can also be deduced from the fact that the graph must intersect the diagonal asymptote.

- Turning point: Solve $f_3'(x) = 0$.

$$f_3'(x) = \frac{(3x^2 - 2x + 3)(x^2+1) - 2x(x^3 - x^2 + 3x + 1)}{(x^2+1)^2} = \frac{x^4 - 4x + 3}{(x^2+1)^2}$$

$$\frac{x^4 - 4x + 3}{(x^2+1)^2} = 0 \Rightarrow x^4 - 4x + 3 = 0 \Rightarrow x = 1$$ (by inspection). $y = f_3(1) = 2$

**Question 2**

Substitute $z = x + iy \quad x, y \in R$ :

$$3\left(x^2 + 2xyi - y^2\right) - 2(x - iy) = \left|2(x + iy) - 1\right|$$

$$\Rightarrow 3x^2 + 6xyi - 3y^2 - 2x + 2yi) = \left|(2x - 1) + 2yi\right|$$

$$\Rightarrow \left(3x^2 - 3y^2 - 2x\right) + i(6xy + 2y) = \left|(2x - 1) + 2yi\right|$$

Real

(1)

Equate real parts of (1): $3x^2 - 3y^2 - 2x = \left|(2x - 1) + 2yi\right|$ (2)

Equate imaginary parts of (1):

$6xy + 2y = 0 \Rightarrow 2y(3x + 1) = 0$ therefore $y = 0$ or $x = -\frac{1}{3}$

| | |
|---|---|
| **Case 1:** $y = 0$.<br>Substitute $y = 0$ into equation (2): $3x^2 - 2x = \left|2x - 1\right|$ | **1 mark** |

**Case 1 (a):** $2x - 1 \geq 0 \Rightarrow x \geq \frac{1}{2}$

$3x^2 - 2x = 2x - 1 \quad \Rightarrow 3x^2 - 4x + 1 = 0 \quad \Rightarrow (3x - 1)(x - 1) = 0 \quad \Rightarrow x = 1$

( $x = \frac{1}{3}$ is rejected because $x \geq \frac{1}{2}$ )

| | |
|---|---|
| **Answer:** $z = 1$ | **1 mark** |

**Case 1 (b):** $2x - 1 < 0 \Rightarrow x < \frac{1}{2}$

$3x^2 - 2x = -(2x - 1) \quad \Rightarrow 3x^2 = 1 \quad \Rightarrow x = -\frac{1}{\sqrt{3}}$

( $x = \frac{1}{\sqrt{3}}$ is rejected because $x < \frac{1}{2}$ ).

| | |
|---|---|
| **Answer:** $z = -\frac{1}{\sqrt{3}}$. | **1 mark** |
| **Case 2:** $x = -\frac{1}{3}$<br>Substitute into equation (2) and simplify:<br>$1 - 3y^2 = \left\lvert -\frac{5}{3} + 2yi \right\rvert \quad \Rightarrow 1 - 3y^2 = \sqrt{\frac{25}{9} + 4y^2}$<br>There is no real solution (proof is not required). | **1 mark** |

The fact that there is no real solution can be seen by inspecting the graphs of $f(y) = 1 - 3y^2$ and $g(y) = \sqrt{\frac{25}{9} + 4y^2}$ and noting that the graphs do not intersect each other.

**Question 3**

$v = \frac{dx}{dt} = 2 - e^{-x} \quad \Rightarrow \frac{dt}{dx} = \frac{1}{2 - e^{-x}}$

| | |
|---|---|
| $\Rightarrow t = \int \frac{1}{2 - e^{-x}}\, dx$ | **1 mark** |

$= \int \frac{e^x}{2e^x - 1}\, dx$

Substitute $u = e^x$:

$t = \int \frac{1}{2u - 1}\, du \quad = \frac{1}{2}\log_e |2u - 1| + c$, where $c \in R$

| | |
|---|---|
| $t = \frac{1}{2}\log_e |2e^x - 1| + c$ | **1 mark** |

$\Rightarrow e^{2(t-c)} = |2e^x - 1| \quad \Rightarrow e^{-2c}e^{2t} = |2e^x - 1| \quad \Rightarrow \pm e^{-2c}e^{2t} = 2e^x - 1$

$\Rightarrow Ae^{2t} = 2e^x - 1$, where $A = \pm e^{-2c}$.

Substitute $x = 0$ when $t = 0$: $A = 1$.

| | |
|---|---|
| $e^{2t} = 2e^x - 1$ | **1 mark** |

Substitute $t = \log_e(7)$:

$2e^x - 1 = e^{2\log_e(7)} = e^{\log_e(49)} = 49$

$\Rightarrow 2e^x = 50 \qquad \Rightarrow e^x = 25$

| | |
|---|---|
| **Answer:** $x = \log_e(25)$<br>Accept $x = 2\log_e(5)$ | **1 mark** |

**Question 4**

The value of $a = \dfrac{d^2x}{dt^2}$ when $t = 1$ and $x = 1$ is required.

Use implicit differentiation:

Differentiate $t^3 + 2x^3 - xt = 2$ with respect to *t*: $3t^2 + \underbrace{6x^2\dfrac{dx}{dt}}_{\text{Chain Rule}} - \underbrace{x - t\dfrac{dx}{dt}}_{\text{Product Rule}} = 0$

| | |
|---|---|
| **Result of implicit differentiation:** $3t^2 + 6x^2\dfrac{dx}{dt} - x - t\dfrac{dx}{dt} = 0$ | **1 mark** |

Substitute $t = 1$ and $x = 1$: $3 + 6\dfrac{dx}{dt} - 1 - \dfrac{dx}{dt} = 0 \qquad \Rightarrow \dfrac{dx}{dt} = -\dfrac{2}{5}$.

Differentiate $3t^2 + 6x^2\dfrac{dx}{dt} - x - t\dfrac{dx}{dt} = 0$ with respect to *t*:

$$6t + \left(12x\frac{dx}{dt}\right)\frac{dx}{dt} + 6x^2\frac{d^2x}{dt^2} - \frac{dx}{dt} - \frac{dx}{dt} - t\frac{d^2x}{dt^2} = 0$$

Chain Rule Product Rule product Rule

| | |
|---|---|
| **Result of implicit differentiation:** $6t + 12x\frac{dx}{dt}\times\frac{dx}{dt} + 6x^2\frac{d^2x}{dt^2} - \frac{dx}{dt} - \frac{dx}{dt} - t\frac{d^2x}{dt^2} = 0$ . | **1 mark** |

Substitute $t = 1$ and $x = 1$ and $\frac{dx}{dt} = -\frac{2}{5}$:

$$6 + 12\left(-\frac{2}{5}\right)\left(-\frac{2}{5}\right) + 6\frac{d^2x}{dt^2} - \left(-\frac{2}{5}\right) - \left(-\frac{2}{5}\right) - \frac{d^2x}{dt^2} = 0 \quad \Rightarrow 6 + \frac{48}{25} + 5\frac{d^2x}{dt^2} + \frac{4}{5} = 0$$

| | |
|---|---|
| **Answer:** $a = -\frac{218}{125}$ m s$^{-2}$ | **1 mark** |

**Question 5**

Use integration by parts to calculate $\int x\tan^{-1}\left(\frac{x}{2}\right)dx$:

$$\int u\,dv = uv - \int v\,du$$

$$u = \tan^{-1}\left(\frac{x}{2}\right) \quad \Rightarrow du = \frac{2}{x^2+4}dx$$

$$dv = x\,dx \quad \Rightarrow v = \frac{1}{2}x^2$$

| | |
|---|---|
| $\int x\tan^{-1}\left(\frac{x}{2}\right)dx = \frac{1}{2}x^2\tan^{-1}\left(\frac{x}{2}\right) - \int \frac{x^2}{x^2+4}\,dx$ | **1 mark** |

$$= \frac{1}{2}x^2\tan^{-1}\left(\frac{x}{2}\right) - \int \frac{(x^2+4)-4}{x^2+4}\,dx \qquad = \frac{1}{2}x^2\tan^{-1}\left(\frac{x}{2}\right) - \int 1 - \frac{4}{x^2+4}\,dx$$

| | |
|---|---|
| $= \frac{1}{2}x^2\tan^{-1}\left(\frac{x}{2}\right) - x + 2\tan^{-1}\left(\frac{x}{2}\right)$ | **1 mark** |

Therefore:

$$\int_0^{2\sqrt{3}} x\tan^{-1}\left(\frac{x}{2}\right)dx = \frac{1}{2}(12)\tan^{-1}\left(\sqrt{3}\right) - 2\sqrt{3} + 2\tan^{-1}\left(\sqrt{3}\right) - 0 \qquad = 6\left(\frac{\pi}{3}\right) - 2\sqrt{3} + 2\left(\frac{\pi}{3}\right)$$

| | |
|---|---|
| **Answer:** $\frac{8\pi}{3} - 2\sqrt{3}$<br>Accept $\frac{8\pi - 6\sqrt{3}}{3}$ | **1 mark** |

**Question 6**

Confidence interval endpoints of a $C$% confidence interval:

$$\bar{x} \pm z_c \frac{\sigma}{\sqrt{n}}$$

where $z_c$ is such that $\Pr(-z_c < Z < z_c) = \Pr(|Z| < z_c) = \frac{C}{100}$

**Part a.:**

Compare the endpoints of the given confidence interval (201.2, 202.6) with $\bar{x} \pm z_c \frac{\sigma}{\sqrt{n}}$

It follows that:

$$202.6 - 201.2 = 2z_c \frac{\sigma}{\sqrt{n}} \quad \Rightarrow 1.4 = 2k \frac{5}{\sqrt{64}} = \frac{5}{4}k \quad \Rightarrow k = \frac{4(1.4)}{5} = \frac{28}{25}$$

| | |
|---|---|
| **Answer:** $k = \frac{28}{25}$<br>Accept 1.12 | **1 mark** |

**Part b.:**

**Method 1:**

Compare the endpoints of the given confidence interval (201.2, 202.6) with $\bar{x} \pm z_c \frac{\sigma}{\sqrt{n}}$

It follows that:

$$202.6 + 201.2 = 2\bar{x} \quad \Rightarrow 403.8 = 2\bar{x} \quad \Rightarrow \bar{x} = 201.9$$

Substitute $\bar{x} = 201.9$, $n = 4 \times 64$, $\sigma = 5$ and $z_c = k = 1.12$ into $\left(\bar{x} - z_c \frac{\sigma}{\sqrt{n}},\ \bar{x} + z_c \frac{\sigma}{\sqrt{n}}\right)$

**Method 2:**

Let $n' = 4n$. Width of new confidence interval:

$$2k \frac{\sigma}{\sqrt{n'}} = 2k \frac{\sigma}{\sqrt{4n}} = \frac{1}{2}\left(\underbrace{2k \frac{\sigma}{\sqrt{n}}}_{\text{Width of old confidence interval}}\right) = \frac{1}{2}(1.4) = 0.7$$

$\bar{x}$ remains the same therefore the new confidence interval is

$$\left(201.9 - \frac{1}{2}(0.7),\ 201.9 + \frac{1}{2}(0.7)\right)$$

| | |
|---|---|
| **Answer:** (201.55, 202.25) | **1 mark** |

**Part c.:**

Let *X* be the random variable "*Mass of a genetically modified apple*".

*X* is normally distributed with a mean of 144 grams and a standard deviation of 8 grams.

$$\Pr\left(X_1 + X_2 < \frac{7}{4}X_3\right) = \Pr\left(X_1 + X_2 - \frac{7}{4}X_3 < 0\right)$$

is required

where $X_1$, $X_2$, $X_3$ are independent copies of $X$.

Let *Y* be the random variable $Y = X_1 + X_2 - \frac{7}{4}X_3$. Then $\Pr(Y < 0)$ is required.

**Note:** Using the random variable $X + X - \frac{7}{4}X = \frac{1}{4}X$ is incorrect: $X_1 + X_2 - \frac{7}{4}X_3 \neq \frac{1}{4}X$

$Y$ follows a normal distribution since $X_1$, $X_2$, $X_3$ are independent normal random variables.

$$\mathrm{E}(Y) = \mu_Y = \mu_{X_1} + \mu_{X_2} - \frac{7}{4}\mu_{X_3} = \frac{1}{4}\mu_X = \frac{1}{4}(144) = 36$$

$$\mathrm{Var}(Y) = 1^2\mathrm{Var}(X_1) + 1^2\mathrm{Var}(X_2) + \left(\frac{7}{4}\right)^2\mathrm{Var}(X_3) = 2\mathrm{Var}(X) + \frac{49}{16}\mathrm{Var}(X) = \frac{81}{16}\mathrm{Var}(X)$$

$$\Rightarrow \mathrm{sd}(Y) = \sigma_Y = \frac{9}{4}\mathrm{sd}(X) = \frac{9}{4}(8) = 18$$

| | |
|---|---|
| $Y = X_1 + X_2 - \frac{7}{4}X_3$<br><br>$(\mu_Y = 36,\ \sigma_Y = 18)$<br>*Y* ~ Normal | **1 mark:**<br><br>Definition.<br><br>Distribution. |

| | |
|---|---|
| $\Pr(Y < 0)$ | Probability. |

$Z = \frac{Y - \mu_Y}{\sigma_Y} = \frac{0 - 36}{18} = -2$ therefore $\Pr(Y < 0) = \Pr(Z < -2)$.

$\Pr(|Z| \le 2) = 0.9545 \Rightarrow \Pr(Z < -2) = \frac{1}{2}(1 - 0.9545) = \frac{1}{2}(0.0455)$.

| | |
|---|---|
| $Z = -2$.<br><br>**Answer:** 0.023 | **1 mark:**<br><br>Calculation of $Z$.<br><br>Answer. |

**Question 7**

$$S = 2\pi \int_0^{\frac{\pi}{2}} x\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Substitute $\frac{dx}{dt} = -3\cos^2(t)\sin(t)$ and $\frac{dy}{dt} = 3\sin^2(t)\cos(t)$:

$$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\cos^4(t)\sin^2(t) + 9\sin^4(t)\cos^2(t) \qquad = 9\cos^2(t)\sin^2(t)\left(\cos^2(t) + \sin^2(t)\right)$$

$$= 9\cos^2(t)\sin^2(t)$$

| | |
|---|---|
| $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = 9\cos^2(t)\sin^2(t)$ | **1 mark** |

| | |
|---|---|
| $S = 2\pi \int_0^{\frac{\pi}{2}} \cos^3(t)\sqrt{9\cos^2(t)\sin^2(t)}\, dt$ | **1 mark** |

$$= 6\pi \int_0^{\frac{\pi}{2}} \cos^3(t)\,|\cos(t)\sin(t)|\, dt \qquad = 6\pi \int_0^{\frac{\pi}{2}} \cos^3(t)\cos(t)\sin(t)\, dt$$

| | |
|---|---|
| Since $\cos(t)\sin(t) \geq 0$ for $0 \leq t \leq \frac{\pi}{2}$<br>$= 6\pi \int_0^{\frac{\pi}{2}} \cos^4(t)\sin(t)\, dt$ | **1 mark:**<br>Justification for dropping modulus.<br>Integral. |

Substitute $u = \cos(t)$:

$$S = -6\pi \int_1^0 u^4\, du \qquad = 6\pi \int_0^1 u^4\, du \qquad = \frac{6\pi}{5}\left[u^5\right]_0^1$$

.

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| **Answer:** $\frac{6\pi}{5}$ square units. | **1 mark** |

**Question 8**

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| • Let $S(n)$ be the conjecture $\frac{(f_1(x)\cdots f_n(x))'}{f_1(x)\cdots f_n(x)}=\frac{f_1'(x)}{f_1(x)}+\ldots+\frac{f_n'(x)}{f_n(x)}$<br>• Check $n=2$:<br>$=\frac{(f_1(x)f_2(x))'}{f_1(x)f_2(x)}=\frac{f_1'(x)f_2(x)+f_1(x)f_2'(x)}{f_1(x)f_2(x)}=\frac{f_1'(x)}{f_1(x)}+\frac{f_2'(x)}{f_2(x)}=$<br>LHS RHS<br>Therefore $S(2)$ is true.<br>• Assume $S(k)$ is true for some $k>2\in N$:<br>$\frac{(f_1(x)\cdots f_k(x))'}{f_1(x)\cdots f_k(x)}=\frac{f_1'(x)}{f_1(x)}+\ldots+\frac{f_k'(x)}{f_k(x)}$ | **1 mark:**<br>Statement.<br>Base case.<br>Inductive hypothesis. |

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| • Show that if $S(k)$ is true then it follows that $S(k+1)$ is true:<br>$\frac{(f_1(x)\cdots f_k(x)f_{k+1}(x))'}{f_1(x)\cdots f_k(x)f_{k+1}(x)}=\frac{([f_1(x)\cdots f_k(x)]f_{k+1}(x))'}{f_1(x)\cdots f_k(x)f_{k+1}(x)}$<br>$=\frac{(f_1(x)\cdots f_k(x))'f_{k+1}(x)+(f_1(x)\cdots f_k(x))f_{k+1}'(x)}{f_1(x)\cdots f_k(x)f_{k+1}(x)}$ | **1 mark:**<br>Statement.<br>Recognition that<br>$f_1(x)\ldots f_k(x)f_{k+1}(x)$<br>is a product of the two functions<br>$f_1(x)\ldots f_k(x)$<br>and |

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| | $f_{k+1}(x)$ . |

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| $=\dfrac{\left(\dfrac{f_1'(x)}{f_1(x)}+\ldots+\dfrac{f_k'(x)}{f_k(x)}\right)(f_1(x)\cdots f_k(x))f_{k+1}(x)+(f_1(x)\cdots f_k(x))f_{k+1}'(x)}{f_1(x)\cdots f_k(x)f_{k+1}(x)}$<br><br>Using the inductive hypothesis<br><br>$=\dfrac{\left(\dfrac{f_1'(x)}{f_1(x)}+\ldots+\dfrac{f_k'(x)}{f_k(x)}\right)f_1(x)\cdots f_k(x)f_{k+1}(x)}{f_1(x)\cdots f_k(x)f_{k+1}(x)}+\dfrac{f_1(x)\cdots f_k(x)f_{k+1}'(x)}{f_1(x)\cdots f_k(x)f_{k+1}(x)}$<br><br>$=\left(\dfrac{f_1'(x)}{f_1(x)}+\ldots+\dfrac{f_k'(x)}{f_k(x)}\right)+\dfrac{f_{k+1}'(x)}{f_{k+1}(x)}$<br><br>Which is equivalent to $S(k+1)$ true.<br><br>• Since $S(2)$ is true and it follows that if $S(k)$ is true then $S(k+1)$ is true, it follows from the principle of mathematical induction that $S(n)$ is true for $n \geq 2$. | **1 mark:**<br><br>Use of inductive hypothesis.<br><br>$S(k) \Rightarrow S(k+1)$<br><br>Conclusion. |

**Question 9**

**Part a.:**

$\Pi_1$ : $\underset{\sim}{n_1} = 3\underset{\sim}{i} - 2\underset{\sim}{j} - m\underset{\sim}{k}$

$\Pi_2$ : $\underset{\sim}{n_2} = \underset{\sim}{i} + m\underset{\sim}{j} + 3m\underset{\sim}{k}$

$\Pi_1$ and $\Pi_2$ are perpendicular when $\underset{\sim}{n_1} \cdot \underset{\sim}{n_2} = 0$ :

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| $3 - 2m - 3m^2 = 0$ | **1 mark** |

$\Rightarrow 3m^2 + 2m - 3 = 0 \qquad \Rightarrow m = \frac{-2 \pm \sqrt{4+36}}{6} = \frac{-2 \pm 2\sqrt{10}}{6} = \frac{-1 \pm \sqrt{10}}{3}$

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| **Answer:** $m = \frac{-1 \pm \sqrt{10}}{3}$ | **1 mark** |

**Part b.:**

$\Pi_1$ : $\underset{\sim}{n_1} = 3\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}$

Parametric equations of the line :

$$x + 1 = \frac{2 - y}{2} = \frac{z - 3}{2}$$

$x + 1 = t \quad \Rightarrow x = -1 + t$

$\frac{2 - y}{2} = t \quad \Rightarrow y = 2 - 2t$

$\frac{z - 3}{2} = t \quad \Rightarrow z = 3 + 2t$

Vector in direction of the line (by inspection of the coefficients of *t* in the parametric equations):

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| $\underset{\sim}{l} = \underset{\sim}{i} - 2\underset{\sim}{j} + 2\underset{\sim}{k}$ | **1 mark:** |

Vector perpendicular to $\underset{\sim}{n_1} = 3\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}$ and $\underset{\sim}{l} = \underset{\sim}{i} - 2\underset{\sim}{j} + 2\underset{\sim}{k}$ :

$$\underset{\sim}{n_1} \times \underset{\sim}{l} = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 3 & -2 & 1 \\ 1 & -2 & 2 \end{vmatrix}$$

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| $= \underset{\sim}{i}\begin{vmatrix} -2 & 1 \\ -2 & 2 \end{vmatrix} - \underset{\sim}{j}\begin{vmatrix} 3 & 1 \\ 1 & 2 \end{vmatrix} + \underset{\sim}{k}\begin{vmatrix} 3 & -2 \\ 1 & -2 \end{vmatrix}$ | **1 mark** |

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| **Answer:** $-2\underset{\sim}{i} - 5\underset{\sim}{j} - 4\underset{\sim}{k}$ . | **1 mark** |

**Question 10**

**Part a.:**

$$\underset{\sim}{r}(t) = \int \sin^2(t)\, dt\, \underset{\sim}{i} + \int \sin(t)\, dt\, \underset{\sim}{k} \qquad \underset{\sim}{r}(t) = \frac{1}{2}\int 1 - \cos(2t)\, dt\, \underset{\sim}{i} + \int \sin(t)\, dt\, \underset{\sim}{k}$$

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| $= \left(\frac{t}{2} - \frac{1}{4}\sin(2t)\right)\underset{\sim}{i} - \cos(t)\underset{\sim}{k} + \underset{\sim}{c}$ | **1 mark** |

where $\underset{\sim}{c}$ is an arbitrary constant vector.

The particle is initially at the point (0, 1, 0) therefore $\underset{\sim}{r}(0) = 0\underset{\sim}{i} + \underset{\sim}{j} + 0\underset{\sim}{k}$.

But

$$\underset{\sim}{r}(0) = \left(0 - \frac{1}{4}\sin(0)\right)\underset{\sim}{i} - \cos(0)\underset{\sim}{k} + \underset{\sim}{c} = -\underset{\sim}{k} + \underset{\sim}{c}$$

$$0\underset{\sim}{i} + \underset{\sim}{j} + 0\underset{\sim}{k} = -\underset{\sim}{k} + \underset{\sim}{c} \qquad \Rightarrow \underset{\sim}{c} = \underset{\sim}{j} + \underset{\sim}{k}$$

Therefore:

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| **Answer:** $\underset{\sim}{r}(t) = \left(\frac{t}{2} - \frac{1}{4}\sin(2t)\right)\underset{\sim}{i} + \underset{\sim}{j} + (1 - \cos(t))\underset{\sim}{k}$ | **1 mark** |

**Part b.:**

The particles are moving in the same direction when their velocity vectors are equal:

$$\underset{\sim}{v}(t) = \sin^2(t)\underset{\sim}{i} + \sin(t)\underset{\sim}{k}$$

Particle 1:

$$\frac{d\underset{\sim}{s}}{dt} = \underset{\sim}{v}_2(t) = -2\sin(2t)\underset{\sim}{i} + (2 - 2\cos(2t))\underset{\sim}{k}$$

Particle 2:

Equate velocity components:

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| $\sin^2(t) = -2\sin(2t)$ (1)<br>$\sin(t) = 2 - 2\cos(2t)$ (2) | **1 mark:**<br>Both equations. |

Simultaneous solutions to equations (1) and (2) are required. From equation (1):

$$\sin^2(t) + 2\sin(2t) = 0 \quad \Rightarrow \sin^2(t) + 4\sin(t)\cos(t) = 0 \quad \Rightarrow \sin(t)(\sin(t) + 4\cos(t)) = 0$$

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| **Case 1** $\sin(t) = 0 \quad \Rightarrow t = n\pi$, $n \in Z^+ \cup \{0\}$. **Note:** $t \geq 0$<br>**Case 2:** $\sin(t) + 4\cos(t) = 0 \quad \Rightarrow \tan(t) = -4$ | **1 mark:**<br>Both cases. |

Substitute **case 1** solutions into (2): $\sin(n\pi) = 2 - 2\cos(2n\pi) \quad \Rightarrow 0 = 0$ true.

Therefore accept **case 1** solutions.

Substitute **case 2** solutions into (2):

$\tan(t) = -4$ therefore $\sin(t) = \pm\frac{4}{\sqrt{17}}$ and $\cos(2t) = 1 - 2\sin^2(t) = 1 - \frac{8}{\sqrt{17}}$

$$\pm\frac{4}{\sqrt{17}} = 2 - 2\left(1 - \frac{8}{\sqrt{17}}\right) = \frac{16}{\sqrt{17}}$$

false.

Therefore reject **case 2**.

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| **Answer:** $t = n\pi$, $n \in Z^{+} \cup \{0\}$ | **1 mark:**<br>Answer.<br>Explicit rejection of case 2. |